Formally, MCP produces a probability estimate of the form

$$p_i^{\text{MCP}} = \text{LLM}_\theta(T_i, N_i \mid p_i^{\text{mkt}}), \tag{5}$$

Here, $\text{LLM}_\theta(\cdot)$ denotes the fixed language model with parameters θ, queried via our MCP prompt template (see Fig. 4), which takes as input the market probability p_i^{mkt} (presented on a 0–100 scale in the prompt) together with the textual context (T_i, N_i) and returns a single numeric probability parsed from the model's output.

Mixture Forecast (MIXMCP). Even with an explicit prior, an LLM can overreact to weak or noisy cues. Following classic work on combining probability forecasts (Genest & Zidek, 1986; Ranjan & Gneiting, 2010) and recent work on combining market baselines with LLM-based adjustments (Alur et al., 2025), we use a convex mixture that anchors on the market

$$p_i^{\text{mixMCP}} = \alpha\, p_i^{\text{mkt}} + (1-\alpha)\, p_i^{\text{MCP}}, \tag{6}$$

Where $\alpha \in [0, 1]$ is fixed across instances. This pooling rule keep the market forecast as a stable baseline while still allowing systematic improvements from evidence based updates.

4 EXPERIMENTS

In this section, we empirically evaluate the effectiveness of the methods defined in Section 3. In Sec. 4.1, we describe the experimental setup, followed by Sec. 4.2 which demonstrates the predictive superiority of our proposed approach.

4.1 EXPERIMENTAL SETUP

Dataset. We evaluate on $N = 856$ Kalshi earnings-call mention markets spanning 50 companies and 70 earnings events (Apr–Dec 2025). Each instance i has a cutoff time t_i set to 7 days before the earnings call. The label $y_i \in \{0, 1\}$ is the Kalshi official resolution based on verbatim keyword presence in the released transcript (see Table 6).

Input Construction. At cutoff t_i, we collect two types of information. For the market signal, we set p_i^{mkt} to the most recent Kalshi YES-contract price observed at or before t_i. For textual evidence, T_i is the prior-quarter earnings-call transcript, and N_i contains up to 100 company-related news items retrieved via SERP API, each formatted with title, snippet, source, and date.

Implementation. All experiments use `GPT-5.1` with structured output on a 0–100 scale (rescaled to $[0, 1]$), without fine-tuning. We sweep $\alpha \in [0, 1]$ and choose $\alpha = 0.7$ (with a plateau around 0.6–0.8) on a held-out split. Therefore, we set $\alpha = 0.7$ for the mixture throughout. To prevent leakage, forecasts are generated using only information available before t_i; the target earnings-call transcript is not observed at inference time.

4.2 RESULTS

Our method, MIXMCP, outperforms the *market probability* baseline. As shown in Table 1, MIXMCP combines a strong market forecast with model-based updates derived from market-conditioned prompting (MCP), achieving a lower Brier score and higher accuracy than the market alone. In the following, we analyze why this combination is effective, examining the role of textual context, the impact of market-conditioned prompting, and the regimes in which these updates are most informative.

Richer context improves probabilistic forecasting performance. We first examine the effect of contextual information without any market-conditioned prompting, using a shared user input template across all variants (see 3). As shown in Table 2, richer context consistently improves forecasting performance. Prior-quarter transcripts yield larger gains than news alone, as they capture recurring themes and language patterns specific to each company's communication style, while combining transcripts and news provides complementary signals by reflecting both stable firm characteristics

Table 1: Forecasting performance of MIXMCP compared to the *market probability* baseline. Arrows indicate the direction of better performance (↓ lower is better, ↑ higher is better).

Prediction	**Brier** (↓)	**ECE** (↓)	**Acc** (↑)	**F1** (↑)
Market Probability	0.1402	**0.0651**	79.8	0.840
MIXMCP	**0.1392**	0.0666	**80.3**	**0.842**

Table 2: Prediction performance measured by Brier score, ECE, accuracy, and F1 score. We compare the effect of providing different types of pre-event information as in-prompt context. Note that W/O PROMPTING corresponds to the 'Market Signal as Plain Text' baseline defined in Section 3.

Prediction	**Brier** (↓)	**ECE** (↓)	**Acc** (↑)	**F1** (↑)
Textual context only				
CONTEXT ($\emptyset$)	0.2635	0.1993	63.1	0.707
CONTEXT (N)	0.2372	0.1387	66.2	0.719
CONTEXT (T)	0.2038	0.1105	70.0	0.733
CONTEXT (T, N)	0.1991	0.0928	70.1	0.734
Textual context with market signal				
W/O PROMPTING$_{\{T,N,M\}}$	**0.1674**	**0.0705**	**74.4**	**0.782**

and recent developments. Providing the *market probability* as an additional input—treated simply as another piece of context within the same template—further improves performance and yields the strongest results among all no-prompting variants. This suggests that the market probability serves as a useful aggregate prior even when incorporated naively, without explicitly instructing the model to reason about or update it.

MCP improves over naive use of market probability. The results above indicate that the market probability can serve as a useful prior when provided as plain context. We now examine whether explicitly instructing the model to reason about this prior leads to further improvements. All variants share the same user input template (see. 3) and differ only in how the market probability is incorporated. Under W/O PROMPTING$_{\{T,N,M\}}$, the model receives transcripts, news, and the market probability as plain context, without any instruction to assess or revise the market signal. As shown in Table 3, this naive inclusion underperforms the market baseline and results in poorer calibration. In contrast, Market-Conditioned Prompting (MCP) explicitly frames the market probability as a prior to be evaluated and updated (see. 4). With this instruction alone, MCP substantially improves calibration (ECE: 0.051 vs. 0.071), reduces Brier score (0.1470 vs. 0.1674), and improves accuracy (78.2 vs. 74.4) relative to W/O PROMPTING$_{\{T,N,M\}}$. These results indicate that the gains arise not from access to the market probability itself, but from conditioning the model to reason about and revise it.

MCP outperforms the market when market confidence is intermediate. To further understand the source of MCP's gains, we analyze disagreement cases between MCP and the *market probability*. Table 4 stratifies these disagreement cases by the market-implied probability and reports which method achieves a lower per-instance Brier score within each bin. Among the 70 instances (8.2% of the data) where the two methods differ in their binary predictions, MCP outperforms the market predominantly when the market signal is uncertain. MCP demonstrates its strongest advantage in mid-confidence regimes, where the market signal is inherently ambiguous. When market probabilities fall in the 50–60% range, MCP achieves lower per-instance Brier scores in 17 out of 30 cases (56.7%), and this advantage becomes even more pronounced in the 60–70% range, where MCP wins 5 out of 8 cases (62.5%). These results show that MCP effectively leverages additional evidence to refine market forecasts precisely when the prior is least decisive.

Figure 2 further supports this interpretation. Across confidence bins, MCP improves overall calibration relative to the market (ECE: 0.051 vs. 0.065), with the largest gains occurring in the mid-confidence region. At higher confidence levels, the market signal is already well-aligned with outcomes, and MCP largely preserves this behavior. Taken together, these results indicate that MCP outperforms the market in mid-confidence cases, resolving ambiguity rather than overturning strong market beliefs.

Table 3: Effect of MCP. We compare providing the *market probability* as plain context with explicitly conditioning the model to reason about and update the market prior. W/O PROMPTING$_{\{T,N,M\}}$ appends the *market probability* as context, while MCP applies market-conditioned prompting.

Prediction	**Brier** (↓)	**ECE** (↓)	**Acc** (↑)	**F1** (↑)
W/O PROMPTING$_{\{T,N,M\}}$	0.1674	0.0705	74.4	0.782
MCP	**0.1470**	**0.0514**	**78.2**	**0.822**

Table 4: Disagreement analysis between the *market probability* baseline and MCP. **Disagree** n denotes instances where the two methods differ in binary prediction. Columns report the count of instances where each method achieves a lower Brier score.

Market Prob. Bin	**Disagree** n	MARKET PROB.	MCP
0–50%	18	13	5
50–60%	30	13	**17**
60–70%	8	3	**5**
70–100%	14	13	1
Total	70	42	28

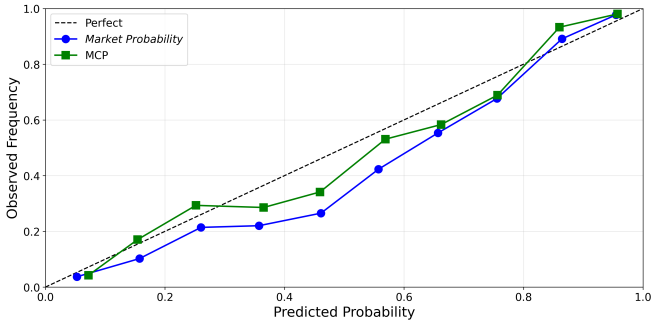

Figure 2: Calibration plot comparing the *market probability* baseline and MCP. The x-axis shows predicted probability and the y-axis shows observed frequency of YES outcomes. Points closer to the diagonal indicate better calibration.

Table 5: Comparison of method variants. W/O PROMPTING provides the *market probability* as plain context, while MCP instructs the model to reason relative to the market baseline. MIXMCP combines the *market probability* and MCP with $\alpha = 0.7$.

Prediction	**Brier** (↓)	**ECE** (↓)	**Acc** (↑)	**F1** (↑)
Market Probability	0.1402	0.0651	79.8	0.840
MCP	0.1470	**0.0514**	78.2	0.822
MIXMCP	**0.1392**	0.0666	**80.3**	**0.842**

MIXMCP: Balanced Integration of Market and Textual Signals. While MCP provides informative corrections in mid-confidence regimes, it does not uniformly outperform the market across all scenarios. This suggests that while LLMs can extract novel insights from text, they may also introduce noise or over-interpret sparse cues. Motivated by this, we employ a convex mixture of MCP with the *market probability*, MixMCP, to systematically balance the established market prior with the LLM's posterior update. As shown in Table 5, MixMCP achieves a lower Brier score (0.1392) compared to both the market (0.1441) and MCP alone (0.1470), while reaching the highest accuracy of 80.3%. In our experiments, an intermediate weighting ($\alpha = 0.7$) yielded the most robust results, indicating that partially dampening the LLM signal improves predictive performance.

We interpret α as a parameter controlling the relative influence of the market signal and the LLM's textual update. From an empirical perspective, α provides a simple mechanism for modulating how strongly the model relies on the market prior versus the LLM output. Viewed this way, the market acts